

needs to post collateral (or equity) worth 10 percent of the underlying asset. A long futures contract allows the investor to buy the asset at a set price in the future. If the market price rises, the investor can buy the asset at a price cheaper than the market price and the profit is the difference between the market price and the contract price. A short position works similarly except that the investor agrees to sell the asset at a fixed price. If the market price falls, the investor can purchase the asset in the open market and sell at the higher price stipulated in the contract. The profit is the difference between the contract price and the market price.

The risk is that the price moves against the investor. For example, if the investor is long with a 10 percent margin and the market price drops by 10 percent, the collateral is gone because the difference between purchasing at the contract price and selling in the open market (at a loss) wipes out the value of the collateral. Importantly, futures are different from options. If the underlying asset's price moves the wrong way in an option contract, the option holder can walk away: the exercise of the option is discretionary – that is why it is called an option – and no trader would exercise an option to guarantee a loss. Futures, however, are obligations. As such, traditional exchanges have mechanisms that seek to minimize the chance the contract holder does not default on a losing position.

The maintenance margin is the main tool to minimize default. Suppose the maintenance margin is 5 percent.